

Regime-Aware Transformers for Equity Return Prediction

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Abstract

*Predicting cross sectional stock returns is difficult because the relationship between technical indicators and future prices shifts with market regimes. A strategy calibrated during low volatility periods may fail entirely when risk conditions tighten. We design, implement, and evaluate three transformer architectures of increasing sophistication for this task. The first is a stock only baseline transformer with no macro input. The second, MacroConcatTransformer, introduces 12 macro indicators by concatenating a learned regime embedding to each token. The third, our **Regime-Aware Transformer** (RAT), replaces concatenation with a conditioning mechanism combining an input level feature gate and Feature wise Linear Modulation (FiLM), both conditioned on the regime embedding. Trained on approximately 395K training samples from S&P 500 constituents, RAT achieves a mean daily Information Coefficient of +0.030 on validation and a gross **Sharpe ratio of 2.68** (Net 2.35 at 20 bps) in long short backtesting. We show that naive macro concatenation fails under cross sectional training due to shift invariance of the ranking loss, and that this is resolved by conditioning mechanisms that break cross stock symmetry.*

1. Introduction

Predicting stock returns from historical price and market data is one of the oldest and hardest problems in quantitative finance. Financial time series are low signal to noise by construction: if a predictive signal were obvious, arbitrage would eliminate it. More fundamentally, the relationships between technical indicators and future returns are non stationary. During periods of market stress, momentum strategies reverse, mean reversion accelerates, and features that predicted returns in calm conditions become misleading. Classical methods respond by fitting regime switching models [4] that explicitly detect market states. Deep learning approaches have largely ignored this structure.

Transformer architectures [11] have demonstrated strong sequence modeling capabilities across language, vision, and

time series [8, 13]. Their self-attention mechanism, however, treats all inputs symmetrically: a transformer trained on mixed regime data must implicitly average regime specific behaviors into a single set of weights, with no mechanism to adapt when market conditions shift.

We build and evaluate three transformer architectures that progressively address this limitation: a **Baseline Transformer** on per stock features only; a **MacroConcatTransformer** that concatenates 12 macroeconomic indicators as a learned regime embedding; and our **Regime-Aware Transformer** (RAT), which replaces concatenation with a *feature gate* and a *FiLM layer* [10], both conditioned on the macro regime embedding $\mathbf{z} \in \mathbb{R}^{32}$.

Contributions. (1) Three trained transformer architectures for cross sectional equity return prediction, each motivated by diagnosing the previous model’s failure. (2) An identification of why macro concatenation fails under ranking based training: shift-invariance of ranking losses renders the concatenated context gradient zero, leaving the regime encoder unlearned. (3) A conditioning mechanism combining a feature gate and FiLM modulation that resolves this failure, achieving $|t| > 2$ significance in long short backtests across all tested rebalancing frequencies.

2. Related Work

Transformers for time series. The Transformer [11] has been adapted for time series forecasting in a series of papers. Informer [13] reduces the quadratic attention cost for long sequences via sparse attention. PatchTST [8] partitions univariate series into patches, dramatically improving long range prediction on benchmark datasets. iTransformer [6] inverts the attention axis, attending across features rather than time steps. These works focus on multistep forecasting of individual series; our task is cross sectional single step return *ranking* across hundreds of assets simultaneously, which requires a ranking based training objective rather than standard regression losses.

Deep learning for stock return prediction. Early neural approaches to stock prediction used LSTMs and GRUs on

univariate price series [3]. More recent work builds cross sectional models that rank stocks jointly. Qlib [12] provides infrastructure for such experiments and includes transformer based baselines. FactorVAE [1] disentangles stock representations into predictive and unpredictable factors in a variational framework. Our work is closest to these cross sectional ranking approaches, but uniquely focuses on conditioning the model on macro context rather than improving the single stock encoder.

Market regime models. Markov switching regression [4] is the classical approach to modeling discrete regime transitions in financial data. More recent work identifies regimes via clustering on macro indicators [9]. A key distinction in our work is that we do *not* define discrete regimes or use labeled transitions. The RegimeEncoder learns a continuous embedding $\mathbf{z}$ end-to-end, jointly optimizing for return prediction.

Feature-wise Linear Modulation. FiLM [10] was proposed for visual reasoning: a language question embedding is used to scale and shift intermediate image feature maps via γ and β vectors. It has since been applied in artistic style transfer [2] and multi modal learning. We adapt FiLM to modulate the pooled transformer representation with a macro regime vector, providing a differentiable and stable conditioning mechanism.

Reversible Instance Normalization. RevIN [5] normalizes each input window by its own statistics and reverses the normalization at output, addressing distribution shift in non stationary time series. We apply RevIN to each 60-day stock window so the transformer sees inputs on a consistent scale regardless of price level or volatility regime.

3. Approach

3.1. Data and Preprocessing

We collect daily OHLCV data for S&P 500 constituents from 2020 to 2025. For each stock on each trading day, we construct a 60-day lookback window of 17 technical features: raw OHLCV, two EMAs (10, 30-day), MACD, ADX, RSI-14, stochastic oscillator, Bollinger Band width, ATR-14, OBV, three return horizons (1d, 5d, 21d), log return, and 21-day realized volatility. Non stationary levels (raw price, volume) are excluded to avoid leakage. Simultaneously, we record 12 macro indicators on each date: VIX, 10-year and 3-month Treasury yields, term spread, VIX z-score (60-day rolling), and 5/21/63-day returns on SPY, HYG, crude oil, gold, and DXY. Features are standardized per-split using training set statistics.

The prediction target is the 5-day forward log return of each stock, **cross-sectionally z-scored** within each trading

date. This makes the labels unit free and focuses the model on ranking stocks rather than predicting absolute return levels. The dataset is split chronologically: training ($\sim 394\text{K}$ samples), validation ($\sim 27\text{K}$), and test ($\sim 131\text{K}$), with no lookahead across splits.

3.2. Model Architecture

The Regime-Aware Transformer (RAT) processes two inputs in parallel: a stock sequence $\mathbf{X} \in \mathbb{R}^{B \times T \times F}$ (batch, 60 time steps, 17 features) and a macro vector $\mathbf{m} \in \mathbb{R}^{B \times 12}$ corresponding to the current trading date. The full forward pass is illustrated in Figure 1.

RegimeEncoder. The macro vector is compressed to a 32-dimensional regime embedding:

$$\mathbf{z} = \text{MLP}(\mathbf{m}) \in \mathbb{R}^{B \times 32} \quad (1)$$

where the MLP has layers $12 \rightarrow 64 \rightarrow 64 \rightarrow 32$ with GELU activations and a LayerNorm after the first linear layer.

Reversible Instance Normalization. Each stock window is normalized by its own per-feature mean and standard deviation:

$$\hat{\mathbf{X}} = \frac{\mathbf{X} - \boldsymbol{\mu}}{\boldsymbol{\sigma} + \epsilon} \cdot \gamma_{\text{rev}} + \beta_{\text{rev}} \quad (2)$$

where $\boldsymbol{\mu}, \boldsymbol{\sigma}$ are computed over the $T = 60$ time dimension. This removes the absolute price level and recent drift from each window, putting all stocks on comparable footing.

Feature Gate. The regime vector conditions which technical features the model emphasizes *before* the transformer sees the sequence:

$$\tilde{\mathbf{X}} = \hat{\mathbf{X}} \odot \sigma(\mathbf{W}_g \mathbf{z} + \mathbf{b}_g) \quad (3)$$

where $\mathbf{W}_g \in \mathbb{R}^{F \times 32}$, σ is the sigmoid, and the gate vector is broadcast over the time dimension. Because each stock has different feature values, the gate produces a distinct scaled output per stock even within a date, preserving gradient flow to the regime encoder. This allows the model to suppress momentum features during high-volatility regimes and emphasize mean-reversion signals, learned entirely from training data without hand-coded regime labels.

Transformer Encoder. The gated sequence is projected to $d_{\text{model}} = 32$ and processed by a single pre-norm transformer encoder layer ($h = 4$ heads, $d_{\text{ff}} = 128$, dropout 0.1) with sinusoidal positional encoding. We use **last-token pooling** (specifically $\mathbf{h} = \mathbf{H}_{:, -1, :}$) rather than mean pooling. Mean pooling dilutes recent timesteps (which carry most of the predictive signal for next-week returns) by a factor of $T = 60$.

FiLM Conditioning. The pooled representation is modulated by the regime vector via a learned affine transform:

$$\mathbf{h}_{\text{cond}} = \mathbf{h} \odot (1 + \gamma(\mathbf{z})) + \beta(\mathbf{z}) \quad (4)$$

where $[\gamma; \beta] = \mathbf{W}_{\text{FiLM}} \mathbf{z} \in \mathbb{R}^{2d_{\text{model}}}$, with $\mathbf{W}_{\text{FiLM}} \in \mathbb{R}^{2d_{\text{model}} \times 32}$ a single linear layer. The FiLM parameters γ and β are learned from the regime embedding $\mathbf{z}$, allowing the model to apply a regime-conditioned scale and shift to the pooled sequence representation.

Prediction Head. The conditioned representation is concatenated with $\mathbf{z}$ and passed through an MLP:

$$\hat{y} = \mathbf{W}_2 \text{GELU}(\mathbf{W}_1[\mathbf{h}_{\text{cond}}; \mathbf{z}] + \mathbf{b}_1) + b_2 \quad (5)$$

with hidden size 64 and dropout 0.1. The concatenation dimension is $d_{\text{model}} + z_{\text{dim}} = 32 + 32 = 64$. Total model parameters: 27,476.

3.3. Training Objective

Standard regression losses (MSE, SmoothL1) are ill suited to cross sectional return prediction because they optimize for absolute magnitude rather than rank ordering. We instead minimize the negative Pearson correlation between predictions and targets within each batch:

$$\mathcal{L} = -\frac{(\hat{\mathbf{y}} - \bar{\hat{\mathbf{y}}}) \cdot (\mathbf{y} - \bar{\mathbf{y}})}{\|\hat{\mathbf{y}} - \bar{\hat{\mathbf{y}}}\| \cdot \|\mathbf{y} - \bar{\mathbf{y}}\| + \epsilon} \quad (6)$$

This is a differentiable proxy for the Information Coefficient (IC), the standard metric in quantitative equity research. The reported validation IC is computed per date using Spearman rank correlation and averaged across dates.

Batching. We train with randomly shuffled mini batches of size 256. Each batch draws samples from across dates, so the IC loss operates as a cross sample ranking objective over mixed date predictions. The targets are pre normalized cross sectionally within each date, so the ranking signal is preserved even when batches span multiple dates.

Optimization. We train with AdamW [7] ($\text{lr} = 10^{-4}$, $\text{wd} = 10^{-2}$), gradient clipping at norm 1.0, and a ReduceLROnPlateau schedule. Training runs for up to 50 epochs with early stopping (patience 10) on validation loss. The best checkpoint (epoch 20) achieves validation IC = +0.030.

3.4. Why Naive Macro Concatenation Fails

A natural baseline (MacroConcatTransformer) repeats $\mathbf{z}$ across all 60 time steps and concatenates it to the stock features before the transformer. This approach fails under ranking based training because the macro vector is identical

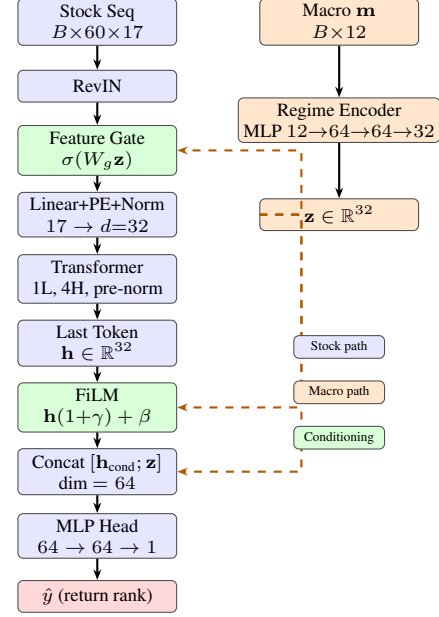


Figure 1. RAT architecture. The macro path (orange) encodes 12 market indicators into a regime vector $\mathbf{z}$, which conditions the stock path (blue) at two points: the *feature gate* before the transformer suppresses irrelevant technical features, and the *FiLM* layer after pooling modulates the sequence summary. Dashed arrows are conditioning connections.

for all stocks on a given date. A ranking loss is shift invariant: adding a constant to all predictions does not change their rank. The macro path therefore contributes zero gradient throughout training, leaving the regime encoder entirely unlearned. RAT sidesteps this by conditioning at two points that break cross stock symmetry: the feature gate multiplicatively scales each stock’s features by different amounts (since each stock has different feature values), and the FiLM layer applies both scale and shift to the sample specific pooled representation. Both produce outputs that differ across stocks within a date, restoring gradient flow to the regime encoder.

4. Experiments and Results

4.1. Model Variants

We train and evaluate all three transformer variants on the same dataset, training procedure, and evaluation protocol described in Section 3.

Baseline Transformer. A pre norm transformer encoder ($d_{\text{model}} = 32$, 1 layer, 4 heads, 15.6K parameters) operating on the 17 stock features only, with no macro input. This model did not converge to above chance IC ($\text{IC} \approx 0$, directional accuracy $\approx 50\%$), establishing that per stock technical features alone carry negligible 5-day predictive signal.

MacroConcatTransformer. Augments the baseline with 12 macro indicators by concatenating a 32-dimensional

regime embedding $\mathbf{z}$ to each input token ($d_{\text{model}} = 64$, 2 layers, 28.4K parameters). Despite the additional information, this model also failed to produce meaningful IC: the ranking loss is shift invariant, so the identical $\mathbf{z}$ shared across all stocks on a given day contributes zero gradient throughout training.

RAT. Replaces concatenation with a conditioning mechanism combining an input level feature gate and FiLM modulation, both driven by $\mathbf{z}$ (27.5K parameters). This is the only configuration in which macro regime information produces non trivial gradients, and the only model to achieve positive IC on the test set.

4.2. Results

Table 1 summarizes IC metrics and Table 2 reports long-short backtest performance. Only RAT reaches positive IC and statistically significant backtest results.

Table 1. IC metrics on the held-out test set. IC is mean daily Spearman rank correlation. Dir. Acc. is fraction of correctly predicted return signs.

Model	Params	Val IC	Test IC	Dir. Acc.
Baseline Transformer	15.6K	≈ 0.000	–	$\approx 50\%$
MacroConcatTransformer	28.4K	≈ 0.000	–	$\approx 50\%$
RAT (ours)	27.5K	+0.030	+0.006	53.4%

Table 2. Long-short backtest results for RAT ($K = 30$). Sharpe ratios annualized net of costs.

Reb.	Cost	Net SR	Ann. Ret	MaxDD	$ t $
21d	0 bps	2.68	+36.9%	−4.7%	3.00
21d	10 bps	2.52	+34.6%	−4.9%	2.81
21d	20 bps	2.35	+32.0%	−5.1%	2.63
5d	0 bps	2.39	+25.9%	−5.4%	2.67
5d	10 bps	1.68	+18.2%	−5.9%	1.88

RAT achieves $|t| > 2$ across all monthly rebalancing configurations and remains profitable at 20 bps cost, a conservative assumption for institutional execution. For reference, SPY buy and hold over the same test period (2024–2025) returned +17% annualized with a Sharpe of 1.61 and a maximum drawdown of −8.0%. RAT exceeds these on all three metrics while being market neutral by construction, with returns generated entirely from cross sectional stock selection rather than passive index exposure.

Figure 2 shows the net equity curves for all three models at $K = 30$, 21-day rebalancing, and 20 bps cost. RAT compounds steadily while both baselines lose money throughout the test period, isolating regime conditioning as the source of the signal.

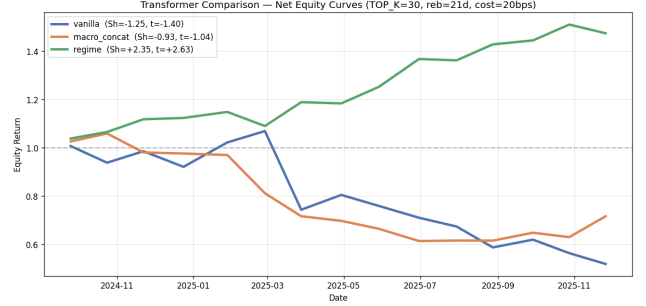


Figure 2. Net equity curves ($K = 30$, 21-day rebalancing, 20 bps cost). RAT (green) compounds positively while both baselines decline. Inset: RAT vs. SPY buy and hold (Sh=1.61, Ann Ret=+17%, DD=−8%); RAT outperforms on all risk adjusted metrics.

4.3. Discussion

The test set daily IC of +0.006 (t -stat +0.75, $N = 313$) is modest but the backtest Sharpe is substantially higher because the long short construction amplifies the per stock signal across 30 positions. The disconnect between IC t -stat (insignificant) and backtest t -stat (significant at +2.63 to +3.00) reflects different variance structures: daily IC has high variance across dates, while portfolio returns average over positions, reducing variance faster than signal.

The three model progression isolates each design choice cleanly. The Baseline Transformer confirms that technical features alone are insufficient. The MacroConcatTransformer shows that macro information is useless if the conditioning mechanism is incompatible with the training objective. RAT demonstrates that the combination of feature gating and FiLM modulation is what allows macro context to influence learned representations and produce statistically significant performance.

4.4. Conclusion

We presented three transformer architectures for cross sectional equity return prediction, each motivated by diagnosing the previous model’s failure. The final architecture, RAT, conditions its computation on a learned macro regime vector through a feature gate and FiLM modulation, achieving a net **Sharpe of 2.35** (Gross 2.68) and annualized returns of +32% net of 20 bps costs over the 2024–2025 test period. Future work should explore two directions: adaptive execution, where the model triggers trades only when its regime confidence exceeds a threshold rather than rebalancing at fixed intervals, and integration of large language models to incorporate unstructured information like earnings call transcripts, central bank communications, and news sentiment directly into the regime embedding, moving toward a fully autonomous trader that combines quantitative signals with natural language understanding of market context.

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